

SDS 8102: Advanced Mathematical Statistics

Practice Problems: Laws of Large Numbers & Empirical Processes

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These are practice problems meant for assessing your own understanding of the topics below. Try them on your own before consulting notes or solutions, and use any difficulty you encounter to identify what to review.

1 A Heavy-Tailed Moving-Average Process

Let $\{Y_t : t \in \mathbb{Z}\}$ be an i.i.d. sequence of Pareto random variables satisfying

$$\mathbb{P}(Y_t > y) = y^{-\alpha}, \quad y \geq 1,$$

where

$$1 < \alpha < 2.$$

Define

$$\varepsilon_t = Y_t - \mathbb{E}[Y_t]$$

and consider the moving-average process

$$X_t = \varepsilon_t + \rho \varepsilon_{t-1}, \quad \rho \in \mathbb{R}.$$

The goal is to show that the sample mean of this dependent heavy-tailed process satisfies a strong law even though the marginal variance is infinite.

1. Show that

$$\mathbb{E}[Y_t] = \frac{\alpha}{\alpha - 1} \quad \text{and} \quad \mathbb{E}[Y_t^2] = \infty.$$

Conclude that

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[|\varepsilon_t|] < \infty, \quad \mathbb{E}[\varepsilon_t^2] = \infty.$$

2. Show that

$$\mathbb{E}[X_t] = 0 \quad \text{and} \quad \mathbb{E}[|X_t|] < \infty.$$

Also show that

$$\mathbb{E}[X_t^2] = \infty.$$

3. Show that $\{X_t : t \in \mathbb{Z}\}$ is 1-dependent; that is, X_t and X_s are independent whenever

$$|t - s| > 1.$$

4. Show that

$$\sum_{t=1}^n X_t = (1 + \rho) \sum_{t=1}^{n-1} \varepsilon_t + \varepsilon_n + \rho \varepsilon_0.$$

5. Use the strong law of large numbers to prove that

$$\frac{1}{n} \sum_{t=1}^{n-1} \varepsilon_t \longrightarrow 0 \quad \text{almost surely.}$$

Show also that

$$\frac{\varepsilon_n}{n} \longrightarrow 0 \quad \text{almost surely.}$$

Hint

If

$$S_n = \sum_{t=1}^n \varepsilon_t,$$

then

$$\frac{\varepsilon_n}{n} = \frac{S_n}{n} - \frac{n-1}{n} \frac{S_{n-1}}{n-1}.$$

Apply the strong law to both terms.

6. Deduce that

$$\frac{1}{n} \sum_{t=1}^n X_t \longrightarrow 0 \quad \text{almost surely.}$$

Explain why a version of the strong law that assumes finite variance cannot be applied directly in this example.

2 The Classical Coupon-Collector Problem

For each $n \geq 2$, let

$$Z_1, Z_2, \dots$$

be i.i.d. random variables uniformly distributed on

$$\{1, \dots, n\}.$$

Define

$$T_n = \inf \{m \geq 1 : \{Z_1, \dots, Z_m\} = \{1, \dots, n\}\}.$$

Thus, T_n is the number of draws required to observe all n possible values.

The goal is to prove that

$$\frac{T_n}{n \log n} \rightarrow 1 \quad \text{in probability.}$$

Thus, approximately $n \log n$ draws are required to collect all n coupons.

1. For $k = 0, 1, \dots, n$, define

$$\tau_k = \inf \{m \geq 0 : \text{card}\{Z_1, \dots, Z_m\} = k\},$$

with $\tau_0 = 0$. Define the waiting times

$$W_{k,n} = \tau_k - \tau_{k-1}, \quad k = 1, \dots, n.$$

Show that

$$T_n = \sum_{k=1}^n W_{k,n}.$$

2. Suppose that $k - 1$ distinct values have already been observed. Show that the probability that the next draw produces a new value is

$$p_{k,n} = \frac{n - k + 1}{n}.$$

Deduce that

$$W_{k,n} \sim \text{Geom}(p_{k,n}),$$

where the geometric distribution is supported on

$$\{1, 2, \dots\}.$$

You may use the fact that

$$W_{1,n}, \dots, W_{n,n}$$

are independent.

3. Using

$$\mathbb{E}[W_{k,n}] = \frac{1}{p_{k,n}},$$

show that

$$\begin{aligned}\mathbb{E}[T_n] &= \sum_{k=1}^n \frac{n}{n-k+1} \\ &= n \sum_{j=1}^n \frac{1}{j}.\end{aligned}$$

Deduce from

$$\frac{1}{\log n} \sum_{j=1}^n \frac{1}{j} \longrightarrow 1$$

that

$$\frac{\mathbb{E}[T_n]}{n \log n} \longrightarrow 1.$$

4. Using

$$\text{Var}(W_{k,n}) = \frac{1 - p_{k,n}}{p_{k,n}^2},$$

show that

$$\begin{aligned}\text{Var}(T_n) &= \sum_{k=1}^n \frac{1 - p_{k,n}}{p_{k,n}^2} \\ &\leq n^2 \sum_{j=1}^n \frac{1}{j^2} \\ &\leq \frac{\pi^2}{6} n^2.\end{aligned}$$

5. Deduce that

$$\text{Var}\left(\frac{T_n}{n \log n}\right) \longrightarrow 0.$$

Use Chebyshev's inequality to prove that

$$\frac{T_n - \mathbb{E}[T_n]}{n \log n} \longrightarrow 0 \quad \text{in probability.}$$

6. Combine the preceding results to conclude that

$$\frac{T_n}{n \log n} \longrightarrow 1 \quad \text{in probability.}$$

Interpret this result in terms of the approximate number of independent draws required to collect all n coupons.

3 A Two-Series Theorem for Positive Random Variables

Let $V_1, V_2, \dots$ be independent nonnegative random variables. Prove that the three-series theorem reduces to a two-series theorem in this setting. Specifically, show that

$$\sum_{n=1}^{\infty} V_n < \infty \quad \text{almost surely}$$

if and only if, for any $c > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(V_n > c) < \infty$$

and

$$\sum_{n=1}^{\infty} \mathbb{E}[V_n \mathbf{1}\{V_n \leq c\}] < \infty.$$

4 Equivalent Criteria for Summability

Let $X_1, X_2, \dots$ be independent nonnegative random variables. Prove that the following statements are equivalent:

- (i) $\sum_{n=1}^{\infty} X_n < \infty \quad \text{almost surely,}$
- (ii) $\sum_{n=1}^{\infty} \mathbb{E} \left[\frac{X_n}{1 + X_n} \right] < \infty,$
- (iii) $\sum_{n=1}^{\infty} \mathbb{E} [\min\{X_n, 1\}] < \infty.$

5 The Strong Law and L^1 Convergence

Let $X_1, X_2, \dots$ be i.i.d. random variables satisfying

$$\mathbb{E}[|X_1|] < \infty,$$

and define

$$S_n = \sum_{k=1}^n X_k.$$

The strong law of large numbers states that

$$\frac{S_n}{n} \longrightarrow \mathbb{E}[X_1] \quad \text{almost surely.}$$

Show also that

$$\boxed{\frac{S_n}{n} \longrightarrow \mathbb{E}[X_1] \quad \text{in } L^1,}$$

or equivalently,

$$\mathbb{E} \left[\left| \frac{S_n}{n} - \mathbb{E}[X_1] \right| \right] \longrightarrow 0.$$

Hint

Prove that the sequence

$$\left\{ \frac{S_n}{n} : n \geq 1 \right\}$$

is uniformly integrable.

6 The Empirical Distribution Function at a Fixed Point

Let $X_1, X_2, \dots$ be i.i.d. random variables with distribution function F . Define the empirical distribution function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}, \quad x \in \mathbb{R}.$$

1. Fix $x \in \mathbb{R}$. Show that

$$\mathbb{E}[\hat{F}_n(x)] = F(x)$$

and

$$\text{Var}(\hat{F}_n(x)) = \frac{F(x)\{1 - F(x)\}}{n}.$$

2. Use the strong law of large numbers to prove that

$$\hat{F}_n(x) \longrightarrow F(x) \quad \text{almost surely}$$

for every fixed $x \in \mathbb{R}$.

3. For $x, y \in \mathbb{R}$, show that

$$\text{Cov}(\hat{F}_n(x), \hat{F}_n(y)) = \frac{F(x \wedge y) - F(x)F(y)}{n}.$$

4. Explain why convergence for each fixed x does not immediately imply

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \longrightarrow 0.$$

7 A Uniform Law over a Finite Class

Let $X_1, \dots, X_n$ be i.i.d. with distribution P , and let

$$\mathcal{F} = \{f_1, \dots, f_M\}$$

be a finite collection of measurable functions satisfying

$$0 \leq f_j(x) \leq 1 \quad \text{for every } x \text{ and } j.$$

Define

$$P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad P f = \mathbb{E}[f(X_1)].$$

1. Use Hoeffding's inequality to show that, for every $f \in \mathcal{F}$,

$$\mathbb{P}(|P_n f - P f| > \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

2. Use the union bound to prove that

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |P_n f - P f| > \varepsilon\right) \leq 2Me^{-2n\varepsilon^2}.$$

3. Deduce that, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \leq \sqrt{\frac{\log(2M/\delta)}{2n}}.$$

4. Suppose now that the class depends on n , with

$$\mathcal{F}_n = \{f_{n,1}, \dots, f_{n,M_n}\}.$$

Give a condition on the growth of M_n under which

$$\sup_{f \in \mathcal{F}_n} |P_n f - P f| \longrightarrow 0 \quad \text{in probability.}$$

8 Rademacher Complexity and Uniform Convergence

Let $X_1, \dots, X_n$ be i.i.d. observations from a distribution P , and let $\mathcal{F}$ be a class of measurable functions

$$f : \mathcal{X} \rightarrow [0, 1].$$

Write

$$P_n f = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad P f = \mathbb{E}[f(X_1)].$$

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher random variables, independent of the data, satisfying

$$\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = \frac{1}{2}.$$

The empirical Rademacher complexity of $\mathcal{F}$ is

$$\widehat{\mathfrak{R}}_n(\mathcal{F}) = \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \mid X_1, \dots, X_n \right].$$

1. Let $X'_1, \dots, X'_n$ be an independent copy of $X_1, \dots, X_n$. Show that

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} |P_n f - P f| \right] \leq \mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - f(X'_i)) \right| \right].$$

2. Use the symmetry of

$$f(X_i) - f(X'_i)$$

and the Rademacher variables to prove the symmetrization inequality

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} |P_n f - P f| \right] \leq 2 \mathbb{E} \left[\hat{\mathfrak{R}}_n(\mathcal{F}_{\pm}) \right],$$

where

$$\mathcal{F}_{\pm} = \mathcal{F} \cup (-\mathcal{F}).$$

3. Now suppose

$$\mathcal{F} = \{f_1, \dots, f_M\}$$

is finite. Conditional on $X_1, \dots, X_n$, show that, for every $\lambda > 0$,

$$\mathbb{E}_{\varepsilon} \left[\exp \left\{ \lambda \sum_{i=1}^n \varepsilon_i f_j(X_i) \right\} \middle| X_1, \dots, X_n \right] \leq \exp \left\{ \frac{\lambda^2}{2} \sum_{i=1}^n f_j(X_i)^2 \right\}.$$

Hint

Use

$$\mathbb{E}[e^{t\varepsilon_i}] = \cosh(t) \leq e^{t^2/2}.$$

4. Deduce Massart's finite-class bound

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \leq \frac{\sqrt{2 \log M}}{n} \max_{1 \leq j \leq M} \left(\sum_{i=1}^n f_j(X_i)^2 \right)^{1/2}.$$

Since $0 \leq f_j \leq 1$, conclude that

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \leq \sqrt{\frac{2 \log M}{n}}.$$

5. Combine the preceding results to obtain

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} |P_n f - P f| \right] \leq 2 \sqrt{\frac{2 \log(2M)}{n}}.$$

6. Use [McDiarmid's inequality](#) to show that, with probability at least $1 - \delta$,

$$\sup_{f \in \mathcal{F}} |P_n f - P f| \leq 2 \sqrt{\frac{2 \log(2M)}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Explain briefly how Rademacher complexity measures the richness of a function class and why a smaller complexity leads to stronger uniform convergence.